

Swami Sahajanand College of Computer Science

B.C.A. SEM-VI

Subject: RDBMS using Oracle-II

UNIT 3

INTRODUCTION TO DBA and DBA Activity

- ◆ Role of DBA.
- ◆ Users: Creating a new user, grant command, deleting user.
- ◆ Privileges: System privileges, object privileges, Assigning object privileges to a user, Viewing User & privileges, revoking a system & an object privileges.
- ◆ Role: Creating a role, Granting privileges & roles to a role, granting role to a user, viewing the role of a user.
- ◆ Database Backup and Recovery
- ◆ Types of Failure
- ◆ Data structure used for Database recovery
- ◆ Import and export

Q-1 What is DBA? Define the Role of DBA (Database Administrator).**❖ What is DBA?**

- ❑ DBA stands for Database Administrator.
- ❑ Database administrator is a person or group of persons under the supervision of a knowledgeable person called Administrator.
- ❑ DBA controls the database structure and sets up the definition for physical as well as logical creation of the database.

❖ Role of DBA/Job of DBA:

- Install and upgrade Oracle and its tools.
- Configure the Oracle instance and SQL * Net.
- Create a database.
- Create, alter, and remove database users and roles.
- Grant and restrict access rights.
- Allocate and manage physical and logical storage structures.
- Develop security strategies.
- Develop backup and recovery procedure.
- Monitor system performance.
- Analyze database performance and implement solutions to problem.
- Communicate with Oracle support service personnel.
- Troubleshoot locking problems.

Q-2 How to create a new user, grant command, deleting user.

- ❑ THE concept of users is very important in RDBMS for database administrators.
- ❑ Anyone who wants to access oracle must be registered as a user of the database, and must have privileges assigned to him or her for the tasks he or she will be able to execute.
- ❑ Roles are groups of privileges that can be assigned to users.

❖ Users:

- One of the most important tasks of an administrator is controlling access to the database.
- Oracle relies on a mechanism that allows you to register a person, called user.
- Each registered user has an access password, which must be provided in various situations.
- Each user is then assigned individual privileges or roles.

• The Create User Command:

- The CREATE USER COMMAND is used for the creation of new users.
- It creates a user or an account through which you can connect to the database.
- You can assign the following optional properties to the user:
 - ◆ Standard tablespace.
 - ◆ Temporary tablespace.
 - ◆ Quotas to allocate space in the tablespace.
 - ◆ Profile containing resource limits.

:: Syntax to create new user::

```
create user <user_name> identified {by password | Externally | Globally as 'cn=user'} [
default tablespace tablespace_name | Temporary tablespace tablespace_name | Quota
{ integer [ K | M ] | Unlimited} on tablespace]...
| profile profile_name | password expire | account { Lock | Unlock }...
```

Syntax: create user <user_name> identified by <password>;

Example: create user manish_dholakiya identified by manish_dholakiya123 default tablespace system profile default;

❖ **To assign privileges (rights) to new user:**

- To assign privileges, you must connect to the administrator using the SYSTEM name and the MANAGER password.
- To connect to the database as a new user, you must use the CONNECT command with following

Syntax: CONNECT user/password

Example: CONNECT manish_dholakiya/manish_dholakiya123

- Both user name & password can be up to 30 characters long.
- If enter a user name that has already been registered, the system will issue an error message and will not allow you to register the new user.
- When user is created you have to assign him or her privileges.
- Before assigning privilege we will try to connect.
- So we have to assign the privileges using the GRANT command.

• **Explain GRANT command:**

- The GRANT command has two distinct forms: one to distribute system privileges and other to distribute object privileges.
- Only DBA can use the GRANT command to distribute system privileges.

- **Syntax:** GRANT system_privileges/ role TO user / role/ public with admin option;
- **Example:** GRANT DBA to manish_dholakiya;

▪ **Privilege:-** The name of the privilege to be assigned.

▪ **User/Role:-** The name of the user or role that is receiving the privileges.

▪ **With admin option:** Allows user or role that receives the privilege to assign it to other users, change it, or even delete it.

❖ **Explain DELETE USER:**

- We can remove a database user with the DROP USER command.
- This command removes both the user and all the objects contained in this user's schema.
- We will need to specify the CASCADE clause of the command.
- To see all users created in system the command is :- **select * from all_users;**

▪ **Syntax:** DROP user <user_name><cascade>;

▪ **Example:** DROP user manish_dholakiya cascade;

Q-3 What is Privileges? Its Types, How to Assigning object privileges to a user, How to View User & privileges and How to revoking a system & an object privileges.

❖ What is a privilege?

- ❑ A privilege is a permission given to the user to access and manipulate a database object in a certain way.
- ❑ For example, one user can be assigned the privilege to select tables but not change them, while another user can read, update and even change the structure of tables and other objects.
- ❑ There are 2 types of privileges: System privilege and Object privilege.

❖ System Privilege:

- A system privilege is the permission to execute certain database actions in a specific type of database object.
- More than 70 types of privileges associated to the actions of a database.
- Name of the privilege is similar to the name of the action it executes.
- For example, a privilege called ALTER TABLE grants a user the right to modify a table.
- List of Some System Privileges:

System Privilege	Type of Action
CREATE	Create an object in user's own schema.
CREATE ANY	Create an object in another user's schema.
CREATE SESSION	Connect to the database.
DROP	Drop any object in user's own schema.
DROP ANY	Drop any object in another user's schema.
ALTER SYSTEM	Manipulate an Instance
ALTER DATABASE	Manipulate the database
ALTER USER	Change user roles or password.
ALTER/ CREATE/ DROP/ MANAGETABLESPACE	Manipulate a tablespace

- **Syntax:** GRANT system_privileges TO user with admin option;
- **Example:** GRANT CREATE SESSION to manish_dholakiya;

❖ Object Privilege:

- An object privilege is the right to perform certain actions in a specific object, such as the right to include a row in a table.
- It is not apply to all the objects of a database.
- Triggers, procedures, indexes, database links and clusters do not have any object privilege.
- List of Object Privileges:

Object Privilege	Object
ALTER	Table & Sequences
DELETE	Tables & Views
EXECUTE	Procedures
INDEX	Tables
INSERT	Tables & Views
REFERENCE	Tables
SELECT	Tables, Views, and Sequences
UPDATE	Tables and Views

• Assigning object privileges to a user/role:

- A variation of the GRANT command allows you to assign object privileges to a user or role.
- The privileges assigned can be: SELECT, ALTER, DELETE, EXECUTE, INSERT, INDEX, REFERENCE, AND UPDATE.
- The following objects can be assigned privileges: tables, views, sequences, snapshots, and synonyms.
- Any user who has authority can grant privileges on tables.

Syntax: GRANT ALL

```
select / Insert / Delete / Index/ Alter /  
Update [(Column_name)] on table_name/ view_name  
TO user/public  
with admin option
```

Example: CONNECT scott/tiger

```
GRANT select on emp to manish_dholakiya;
```

- So now emp table is also accessed by manish_dholakiya.
- Because owner of emp table scott gives privilege to the user manish_dholakiya.
- But manish_dholakiya can only see the data he could not perform any insert, update, delete, etc. operation.
- **Select:**-To select data in a table or view.
- **Insert:**-To insert rows in a table or view.
- **Delete:**-To eliminate rows in a table or view.
- **Update:**-To update a table and, optionally, update only the specified columns.
- **Index:**-To create or eliminate the indexes of a table.
- **Alter:**-To modify a table.
- **All:**-To perform all the above privileges.
- **Table_name:**- Name of the existing tables in the database.
- **View_name:**- Name of the existing views in the database.
- **Column_Name:**- This is optional & determines the columns of tables or views specified in the ON clause. The name of the columns must not be qualified.
- **User:**-Name of the user to whom the privilege is assigned.
- **Public:**-Means that all the current and new users have the privileges specified for the table or view.
- **With Admin Option:** - Allows the user or role that receives the privilege to grant it to other users, modify it, or even delete it.

❖ Revoking a System Privileges:

- We can use REVOKE command to revoke any system privileges.
- System privilege must have been granted with the Admin Option.

```
Example: REVOKE session from manish_dholakiya;
```

❖ Revoking an Object privileges:

- To revoke privilege we have to connect as a user who has the authority to grant privilege.
- A user can not grant or revoke privileges to himself or herself.

```
Example: connect scott/tiger  
revoke select on emp from manish_dholakiya;
```

- So now privilege has been revoked, manish_dholakiya cannot access the table anymore.

Q-4 Define the Role: Creating a role, Grating privileges & roles to a role, Granting role to a user, viewing the role of a user.**❖ What is Role?**

- ❑ Role is a group of privileges assigned to a particular user name.
- ❑ Instead of granting eight privileges to a user, we can create a role that is assigned those eight privileges to the user.
- ❑ Role simplifies the administration of users.
- ❑ Oracle has several predefined roles. Such as....
 - CONNECT
 - RESOURCE
 - DBA
 - EXP_FULL_DATABASE
 - IMP_FULL_DATABASE
- ❑ We can also create user defined roles that meet the needs of our application.
 - **CONNECT :**
 - It allows access permission to the database object.
 - If user is created with CONNECT role then we can select, update or deletes the table rows.
 - User cannot create the object as well as destroy.
 - **RESOURCE:**
 - Allows access to the database.
 - Allows the creation of tables, sequences, procedures, triggers, indexes etc.
 - It allows DDL & DML command operations on the object.
 - Most of the users have RESOURCE role.
 - **DBA:**
 - Provides all the system privileges.
 - It allows granting privileges to other users.
 - DBA user can create & drop any user.
- ❑ Default Oracle users, password & their roles

User Name	Password	Role
SCOTT	TIGER	CONNECT & RESOURCE
SYSTEM	MANAGER	DBA
SYS	SYS1	CONNECT, RESOURCE, DBA, EXP_FULL_DATABASE, IMP_FULL_DATABASE
DEMO	DEMO	CONNECT & RESOURCE

❖ How we can create a Role?**▪ Syntax:****CREATE ROLE Role_name****[NOT IDENTIFIED | IDENTIFIED { BY password | EXTERNALLY | GLOBALLY }];**

- **Role_name:**
 - Name of the role to be crated.
- **Not Identified:**
 - Indicates that this role is authorized by the database.
 - No password is required to enable it.
- **Identified:**
 - Indicates that the user must be authorized by the specified

method before the role can be enabled with the SET ROLE command.

- **BY password:**
 - User must specify the password for Oracle to enable the role.
- **Externally:**
 - Indicates that the user must be authorized by an external service such as operating system or a third party software before the role can be enabled.
- **Globally:**
 - Indicates that the user must have authorization from oracle security service to use the role before it can be enabled with the SET ROLE command, or at the login.

▪ **Example:** CREATE ROLE basic1 identified by manish_dholakiya123;

❖ Granting Privileges and Roles to a role:

- After creating role, we can grant it privileges using the GRANT command.
- **Syntax:**
GRANT role/Name_of_privilege TO user/role/ PUBLIC WITH ADMIN OPTION;
- **Example:** GRANT DBA to basic1;
 - Above example grants the DBA role to basic1.

❖ Granting a Role to a User:

- Now we assign the basic1 role to user manish_dholakiya.
- **Example:** connect manish_dholakiya/manish_dholakiya123;
GRANT basic1 to manish_dholakiya;

❖ Viewing the roles of a user:

- Oracle has some special views that control the privileges and roles.
- One of these views is called USER_ROLE_PRIVS.
- It displays the roles assigned to the current user. Such as....
- **Example:** select * from USER_ROLE_PRIVS;

❖ Deleting a Role:

- DROP ROLE command is used to delete a role.
- User must have the system privilege DROP ANY ROLE to remove a role.
- **Syntax:** DROP ROLE <Role_name>;
- **Example:** DROP ROLE basic1;

Q-5 What is Database Backup and Recovery?**❖ Database Backup:**

- ❑ A backup is a copy of data.
- ❑ This copy can include important parts of the database, such as the control file and data files.
- ❑ A backup is a safeguard against unexpected data loss and application errors.
- ❑ If you lose the original data, then you can reconstruct it by using a backup. Backups are divided into physical backups and logical backups.
- ❑ Physical backups, which are the primary concern in a backup and recovery strategy, are copies of physical database files.
- ❑ You can make physical backups with either the Recovery Manager (RMAN) utility or operating system utilities.
- ❑ In contrast, logical backups contain logical data (for example, tables and stored procedures) extracted with an Oracle helpfulness and stored in a binary file. You can use logical backups to supplement physical backups.
- ❑ Since one or more files can be physically damaged due to a disk failure, media recovery requires the restoration of these files from the most recent backup of the operating system to a database.
- ❑ Database backup is the activity of copying database to a permanent storage area so that they will be preserved in case of equipment failure or other disasters.
- ❑ Equipment failures include system crashes, disk damage etc. Disasters include fire, flood or earthquake.

(1) System Backup:

- A backup of the whole database is a backup of the operating system for all the data files, online redo log files, and control files that make up the oracle database.
- Complete backups are performed when the database is closed and unavailable to users.

(2) Partial Backup:

- A partial backup is a backup made by the operating system as part of the database.
- The backup of the datafiles in an individual tablespace, or the backups of a control file are examples of partial backups.
- Partial backups are useful only when the redo log of the database is being operated in the ARCHIVELOG mode.
- A variety of partial backups can be made to accommodate the backup strategy.
- You can perform a backup of datafiles and control files when a database is open or closed, or when a specific tablespace is online or offline.
- Since the redo log is operated in the ARCHIVELOG mode, the additional backups of the redo log are not necessary; the written redo log is a backup of the completed online redo files.

❖ Database Recovery:

- ❑ Recovery is used to restore a physical backup of a data files or control file.
- ❑ Recovery is used to reconstruct it and make it available to the Oracle database server.

Recovery is used to recover data files in case of failure, recovery also update database applying archived redo logs and online redo logs, this two log files are used to save changes made to the database after the backup was taken.

Q-6 Explain Types of Failure?

- All electronic devices are subject to failures in one form or another. Failures in DBMS can be
- due to transaction, system or media failures etc. Most common types of failures are....

1) User error:

- Sometimes user executes the wrong command on the database this type of error occurs.
- Such as user want to update all the 'CLERK' salary with RS.500 on emp table but during command line he forget to give where clause & condition.
- So all the records are updated & it creates one type of failure because of users wrong input.

2) System crash:

- A hardware, software or network error can occur in the computer system during transaction execution.

3) Transaction error:

- Some erroneous operation, such as integer overflow or division by zero, etc... in the transaction may cause a failure in between.

4) Process Failure:

- A process failure is a failure in a user, server, or background process of a database instance (for example, an abnormal disconnects or process termination).
- When a process failure occurs, the failed subordinate process cannot continue work,
- The Oracle background process PMON detects aborted Oracle processes. If the aborted process is a user or server process, PMON resolves the failure by rolling back the current transaction of the aborted process and releasing any resources that this process was using.

5) Database Instance Failure:

- Database instance failure occurs when a problem arises that prevents an Oracle database instance (SGA and background processes) from continuing to work.
- An instance failure can result from a hardware problem, such as a power outage, or a software problem, such as an operating system crash.
- When using the Oracle Parallel Server, other instances perform instance recovery. For additional information about instance recovery.

6) Disk failure:

- Some disk blocks may lose their data because of a read or write malfunction or because of a disk read/write head error.

7) Natural error:

- This refers to the number of problems including power failure, fire, theft, overwriting tapes or disks by mistake.

Q-7 Data Structure used for Database recovery:

- ❏ Oracle uses several structures to provide for the complete recovery of an instance or a disk failure. These include the **redo log, control files, rollback segment and database backups.sc**

❖ The Redo Log:

- ❏ The redo log is a set of files that protects changed data in a database, it contains changes that are located in the memory and have not yet been written to the data files. Redo logs have two parts

1. **The online redo log.**
2. **The written redo log**

(1) The Online Redo Log:

- The online redo log is a set of two or more files in this log that makes all the changes made to the database, including both uncommitted changes.
- The redo entries are temporarily stored in the redo log buffers of the system global area, and the background process LGWR continuously writes a commit log whenever a user process is submitted to a transaction.
- The online redo log files are used cyclically; for example, when the online redo log contains two files, the first file is filled, and then the second one. Then the first file is reused, as is the second file, and so on. When a file is filled, it receives a number of log sequences to identify the set of redo entries.
- To avoid losing the database because of a single failure point. Oracle can maintain several sets of redo files called multiplexed redo log.
- Copies of the online redo logs that are physically located in separate disks from the multiplexed redo logs.

(2) The Written Redo Log:

- Optionally, the online redo logs that are filled can be written before they are used, creating a written redo log. The offline written redo log files constitute the written redo log.
- The presence or absence of a written redo log is determined by how the redo log is being used. In ARCHIVELOG mode, the online redo log files are written before they are used in the cycle. In NOARCHIVELOG mode, the online redo log files are not written.
- In ARCHIVELOG mode, the database can be completely recovered from the instance or disk failure. It is also possible to make a backup while it is open and available to the user.
- However, to update the written redo log, some additional administrative operations are necessary.

❖ Control Files:

- ❏ The control files of a database maintain, among other things, information about the file structure and the current number of log sequences that are being written to the LGWR.
- ❏ During the normal recovery procedure, information in a control file is used to guide the automated progression of the recovery.
 - **Multiplexed Control File:**
 - Multiplexed control files are a resource similar to the multiplexed redo log. Oracle can simultaneously update several identification control files.

❖ Rollback Segment:

- ❑ The rollback segments log rollback information required by the different functions of oracle. During the database recovery that occurs after the application of all the changes written to the redo log.
- ❑ Oracle uses the information in the rollback segment to undo all the un-submitted transactions. Since the rollbacks are stored in the database buffers, the redo log automatically protects this important recovery information.

❖ Database Backup:

- ❑ Since one or more files can be physically damaged due to a disk failure, media recovery requires the restoration of these files from the most recent backup of the operating system to a database.
- ❑ Database backup is the activity of copying database to a permanent storage area so that they will be preserved in case of equipment failure or other disasters.
- ❑ Equipment failures include system crashes, disk damage etc. Disasters include fire, flood or earthquake.
- ❑ There are several ways to do backups.

(1) System Backup:

- A backup of the whole database is a backup of the operating system for all the data files, online redo log files, and control files that make up the oracle database.
- Complete backups are performed when the database is closed and unavailable to users.

(2) Partial Backup:

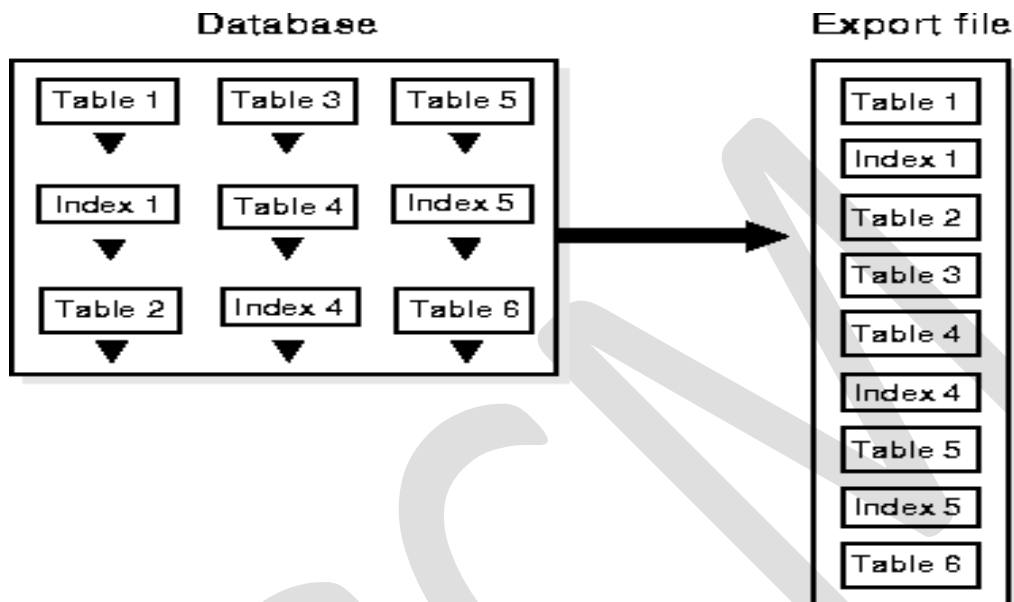
- A partial backup is a backup made by the operating system as part of the database.
- The backup of the datafiles in an individual tablespace, or the backups of a control file are examples of partial backups.
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- A variety of partial backups can be made to accommodate the backup strategy.
- You can perform a backup of datafiles and control files when a database is open or closed, or when a specific tablespace is online or offline.
- Since the redo log is operated in the ARCHIVELOG mode, the additional backups of the redo log are not necessary; the written redo log is a backup of the completed online redo files.

Q-8 Explain Import and Export in detail.

- ❑ Oracle's export (exp) and import (imp) utilities are used to perform logical database backup and recovery.
- ❑ When one export, database objects, it is dumped to a binary file which can then be imported into another Oracle database.
- ❑ These utilities can be used to move data between different machines, databases or schema. However, as they use a proprietary binary file format, they can only be used between Oracle databases.
- ❑ One cannot export data and expect to import it into a non-Oracle database.
- ❑ Various parameters are available to control what objects are exported or imported.
- ❑ To get a list of available parameters, run the exp or imp utilities with the help=yes parameter.

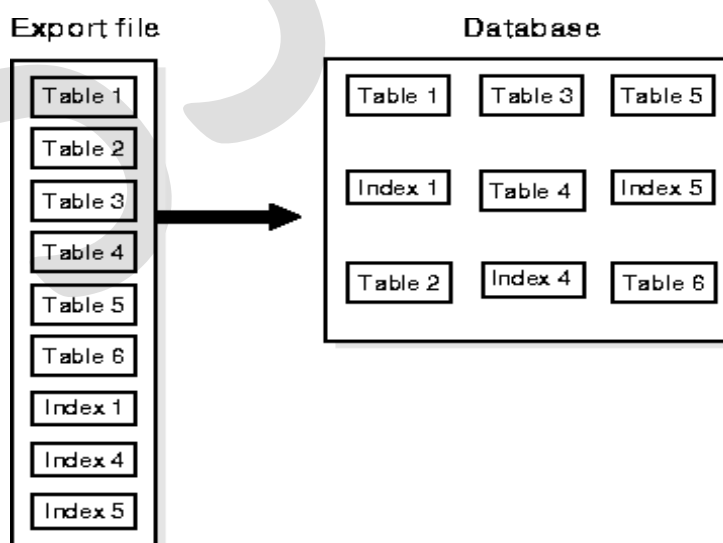
❖ What Is the Export Utility?

- ❑ Export provides a simple way to transfer data objects between Oracle databases, even if they belong to platforms with different hardware and software configurations.
- ❑ Export extracts the object definitions and table data from an Oracle database and stores them in an Oracle binary-format.
- ❑ Export dump to binary file which is typically located on disk or tape.



❖ What Is the Import Utility?

- ❑ The basic concept behind Import is very simple.
- ❑ Import inserts the data objects extracted from one Oracle database by the Export utility (and stored in an Export dump file) into another Oracle database.
- ❑ Export dump files can only be read by Import.
- ❑ Import reads the object definitions and table data that the Export utility extracted from an Oracle database and stored in an Oracle binary-format Export dump file located typically on disk or tape.



- **Order of Import:**

- Table objects are imported as they are read from the export file. The export file contains objects in the following order:

1. Type definitions.
2. Table definitions.
3. Table data.
4. Table indexes.
5. Integrity constraints, views, procedures, and triggers.
6. Bitmap, functional, and domain indexes.

- **How does one use the import/export utilities? :**

- To use imp or exp utilities one must start MS-DOS.
- Then select directory using CD command to take backup.
- The following examples demonstrate how the imp/exp utilities can be used:

```
exp scott/tiger file=emp.dmp log=emp.log tables=emp rows=yes indexes=no
```

```
exp scott/tiger file=emp.dmp tables=(emp,dept)
```

```
imp scott/tiger file=emp.dmp full=yes imp
```

```
scott/tiger file=emp.dmp tables=dept
```